Model Complexity & Investor Learning

MFE5340: AI in FE: Quantitative Investment

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1 Ridgeless Least Squares

- Ridgeless Least Squares
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- Traditional econometrics relies on the principle of simplicity (*parsimony*).

  "Use the fewest parameters necessary to explain a phenomenon."

- Modern machine learning adopts highly parameterized models.
  - GPT-3: 175 billion parameters (Brown et al., 2020).
  - Gu, Kelly and Xiu (2020): Neural networks with 30,000 parameters for return prediction.

- Large models with overparameterization often outperform simple models.

- **Why do highly complex models generalize well in practice?**

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- Traditional view: Balance between bias and variance.

$$\text{Prediction Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

- Complex models:
  - Reduce bias by better approximating the true data-generating process (DGP).
  - Increase variance due to high parameterization.

- Modern findings: Regularization can control variance even in overparameterized models.

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- What is Overparameterization?
  - Models with more parameters than data points ($p > n$).
  - Example: Neural networks with millions of parameters trained on smaller datasets.

- Key Phenomena:
  - Interpolation: Achieving zero training error.
  - Double Descent: A surprising risk curve where test error decreases again as model complexity increases (Hastie et al. 2022).

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- Data Model:

$$y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \epsilon_i, \quad i = 1, \ldots, n$$

where $\boldsymbol{x}_i$ are features in $\mathbb{R}^p$, $\boldsymbol{\beta}$ are true coefficients, and $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ is the random noise.

- Focus on **minimum ℓ_2 norm interpolator**:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{\beta}\|_2 \quad \text{s.t.} \quad \mathbf{X}\boldsymbol{\beta} = \boldsymbol{y} \quad \text{(interpolation)}$$

- This is equivalent to **ridgeless least squares**:

$$\hat{\boldsymbol{\beta}} = \lim_{\lambda \to 0} \arg\min_{\boldsymbol{\beta}} \|\mathbf{X}\boldsymbol{\beta} - \boldsymbol{y}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2.$$

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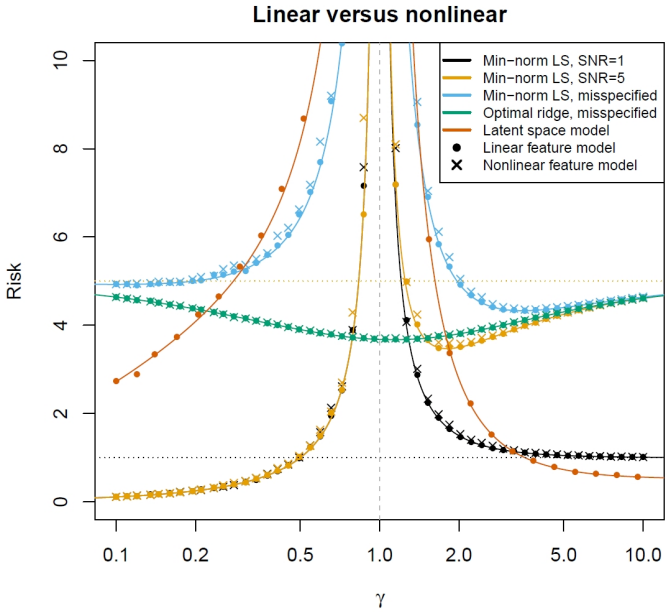
- **Double Descent Risk Curve:**
  - In the underparameterized regime ($p < n$), adding parameters increases test error (classic bias-variance tradeoff).
  - In the overparameterized regime ($p > n$), adding parameters can **reduce test error**.

- Why?
  - Min-norm solution **implicitly regularizes**.
  - Variance decreases in the overparameterized regime.

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Case 1: Isotropic Features

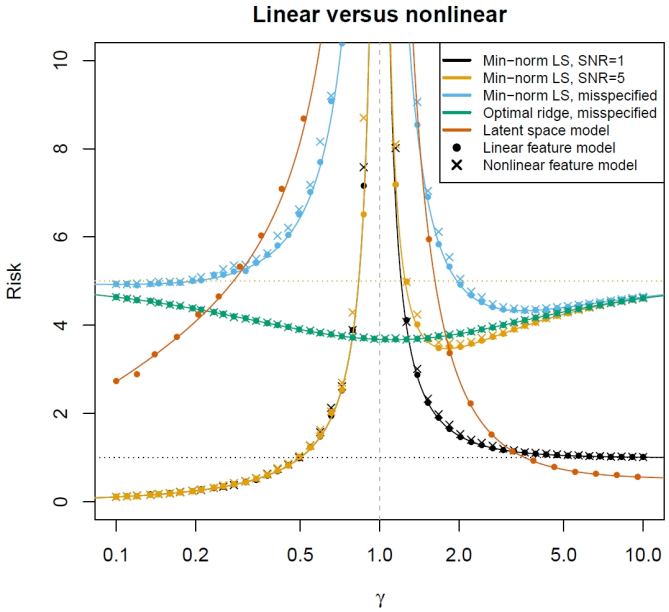
- Features $\boldsymbol{x}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$ (independent, identically distributed).
- Signal-to-noise ratio (SNR): $\text{SNR} = \|\beta\|_2^2/\sigma^2$.
- Test risk depends only on SNR and the ratio $\gamma = p/n$.
- In the overparameterized regime ($\gamma > 1$), risk decreases as γ increases.



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Case 2: Anisotropic Features

- Features have covariance $\boldsymbol{\Sigma} \neq \mathbf{I}_p$.
- $\boldsymbol{\beta}$ may align with top eigenvectors of $\boldsymbol{\Sigma}$.
- Risk depends on the alignment between $\boldsymbol{\beta}$ and $\boldsymbol{\Sigma}$'s eigenvectors.
- Strong alignment with top eigenvectors reduces risk further in the overparameterized regime.



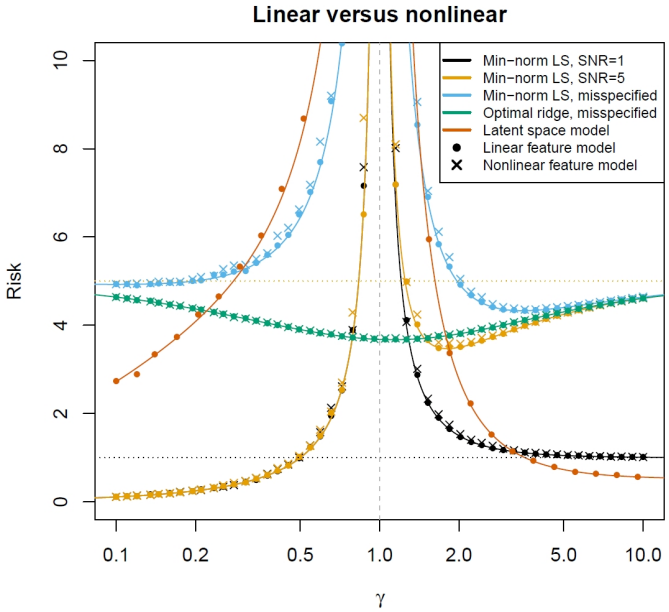
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Case 3: Nonlinear Features

- Features generated by a random one-layer neural network:

$$\boldsymbol{x}_i = \phi(\mathbf{W}\boldsymbol{z}_i), \quad \boldsymbol{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$$

- ϕ: Nonlinear activation function.
- Risk behavior is similar to linear models.
- Insights from simple models can apply to complex settings like neural networks.



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Summary

- "More parameters" does not always mean "overfitting".
- Overparameterized models can generalize well due to implicit regularization.
- Double descent is a universal phenomenon across different settings.
- Insights from simple models (e.g., linear regression) extend to complex models (e.g., neural networks).

High-dimensional models challenge traditional statistical wisdom. But, how do these results extend to real-world financial datasets?

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- True return model:

$$R_{t+1} = f(\boldsymbol{x}_t) + \varepsilon_{t+1},$$

where $\boldsymbol{x}_t$ are covariates (known to the analyst), f is a unknown function mapping covariates to expected returns, and ε_{t+1} are i.i.d errors.

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Return Prediction Problem

- True return model:

$$R_{t+1} = f(\boldsymbol{x}_t) + \varepsilon_{t+1},$$

where $\boldsymbol{x}_t$ are covariates (known to the analyst), f is a unknown function mapping covariates to expected returns, and ε_{t+1} are i.i.d errors.

- The analyst uses a neural network to approximate f:

$$f(\boldsymbol{x}_t) \approx \sum_{i=1}^{p} s_{i,t}\beta_i$$

where $s_{i,t}$ are nonlinear transformations of the original features, i.e., $s_{i,t} = \tilde{f}(\boldsymbol{w}_i^\top \boldsymbol{x}_t)$, $\boldsymbol{w}_i$ are weights of the neural network, and β_i are parameters to be estimated.

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- Empirical model:

$$R_{t+1} = \sum_{i=1}^{p} s_{i,t}\beta_i + \tilde{\varepsilon}_{t+1}.$$

- How to determine the optimal number of predictors p, i.e., the model complexity?
- Kelly et al. (2024): **Use the most complex model feasible within computational constraints**.
  - Out-of-sample prediction accuracy and portfolio performance improve with increasing model complexity.

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Empirical Analysis of Kelly et al. (2024)

- Focus on market timing in the US stock market, i.e., forecasting aggregate market return.
- Info set consists of 15 predictor variables from Welch and Goyal (2008):

Variable	Meaning	Variable	Meaning	Variable	Meaning
d/p	Dividend Price Ratio	d/y	Dividend Yield	e/p	Earnings Price Ratio
d/e	Dividend Payout Ratio	svar	Stock Variance	let	Long Term Rate of Returns
b/m	Book-to-Market Ratio	ntis	Net Equity Expansion	eqis	Percent Equity Issuing
tbl	Treasury Bills	lty	Long Term Yield	tms	The Term Spread
dfr	Default Return Spread	infl	Inflation	lag-1	one lag of the market return

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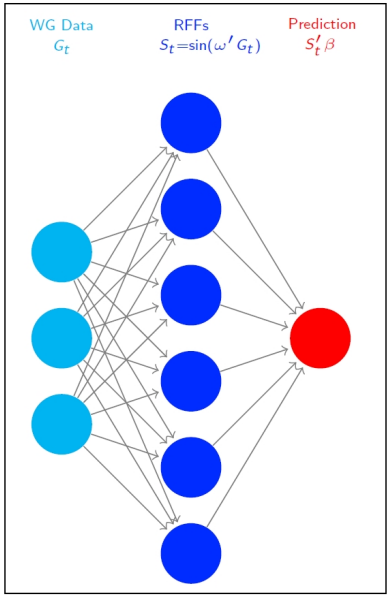
- Recall the empirical model $R_{t+1} = \boldsymbol{s}_t^\top \boldsymbol{\beta} + \tilde{\varepsilon}_{t+1}$.
- Question: How to construct $\boldsymbol{s}_t$ efficiently while capturing sufficient complexity? Answer: Random Fourier Features (RFF).
- Transform 15×1 vector $\boldsymbol{x}_t$ using random weights $\boldsymbol{w}_i$:

$$s_{i,t} = \sin(\boldsymbol{w}_i^\top \boldsymbol{x}_t), \quad \boldsymbol{w}_i \sim \text{iid } \mathcal{N}(\mathbf{0}, \gamma \mathbf{I}),$$

  where $s_{i,t}$ represents a random linear combination of $\boldsymbol{x}_t$ passed through a non-linear activation.
- RFF provides a flexible framework to approximate complex functions with simple methods. Scales from low-dimensional (e.g. $p = 2$) to high-dimensional (e.g., $p > 10,000$) models.

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- Two-Layer Neural Network Interpretation:
  - First Layer: Fixed random weights ($\boldsymbol{w}_i$).
  - Second Layer: Optimized weights (regression $\boldsymbol{\beta}$).
- RFF provides a simple yet powerful way to approximate complex functions using random projections.



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- Setting:
  - Training window: $T = 12$ (one year).
  - Vary feature set size $p = 2, \ldots, 12,000$.
  - Apply shrinkage parameter $\log_{10}(\lambda)$ ranging from -3 to 3.

- Steps:
  - Generate 12,000 RFFs.
  - Fix model parameters (p, λ).
  - Perform recursive out-of-sample (OOS) prediction and timing strategy.
  - Calculate expected return, volatility, and Sharpe ratio.

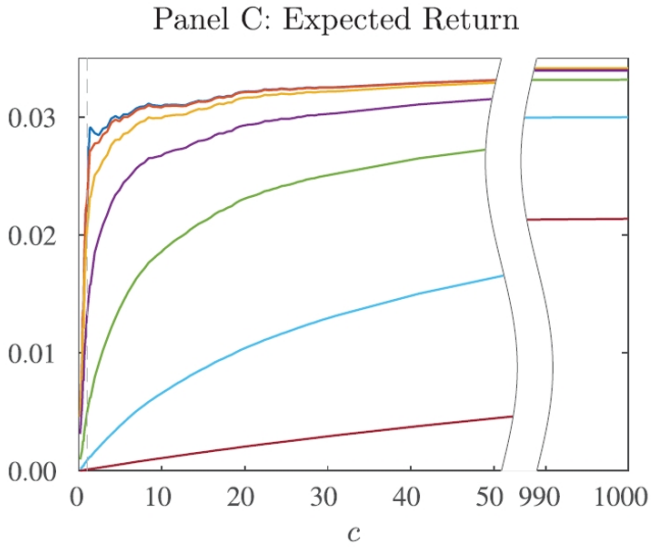
- **BUT...**

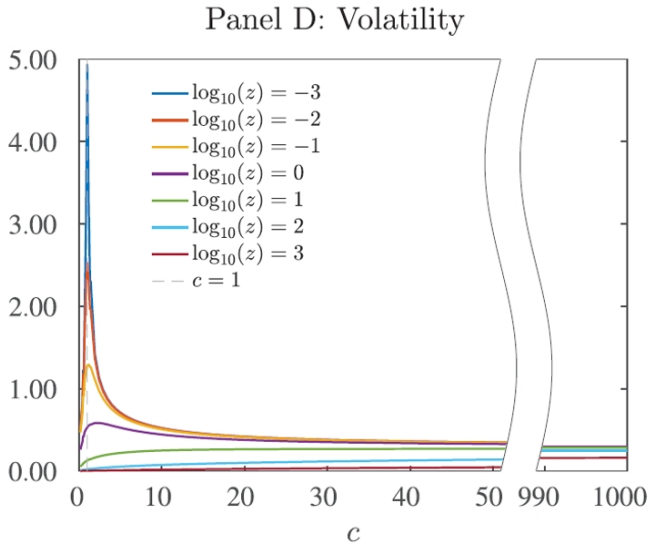
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- RFF has an inherent randomness nature, which means...

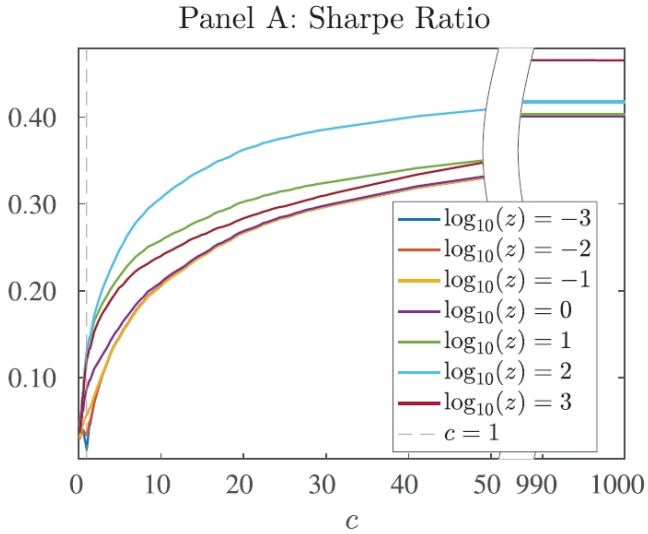
> The inherent randomness of RFFs means that estimates of out-of-sample performance tend to be noisy for models with low P. We therefore repeat the analysis steps from (i) to (iv) 1,000 times with independent draws of the RFFs, and then average the performance statistics across repetitions.

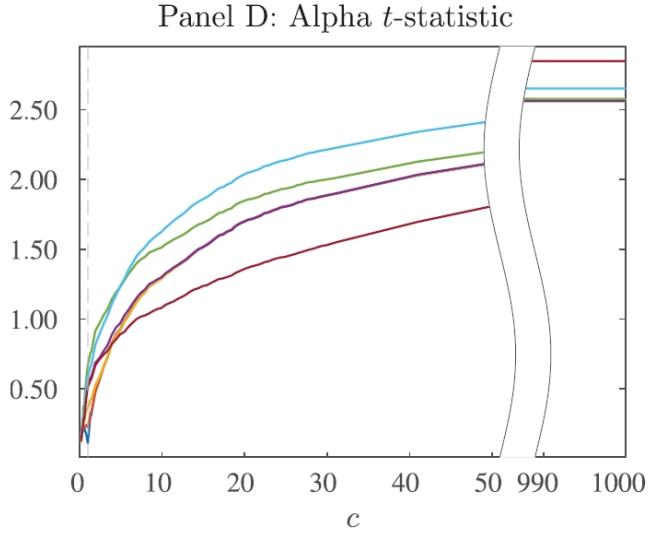
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Martin and Nagel (2022) in a Nutshell

- Investors face a **high-dimensional** prediction problem when valuing cash flows, dealing with numerous covariates. Traditional models assume **rational expectations**, ignoring the complexity of real-world learning.

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Martin and Nagel (2022) in a Nutshell

- Investors face a **high-dimensional** prediction problem when valuing cash flows, dealing with numerous covariates. Traditional models assume **rational expectations**, ignoring the complexity of real-world learning.

- **Learning errors introduce systematic biases into asset prices, even in the absence of risk premia or behavioral biases.** This leads to:

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  — In-Sample: Return predictability arises due to investor learning errors. High-dimensional settings amplify this effect, making returns appear predictable to econometrics (i.e., you! the quant!) *ex post*.

  — Out-of-Sample: No out-of-sample predictability exists for real-world investors (another version of you, but in live-trading env!) *ex ante*.

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  - Out-of-Sample: No out-of-sample predictability exists for real-world investors (another version of you, but in live-trading env!) *ex ante*.
- The proliferation of return predictors ("factor zoo") may reflect the high-dimensional learning challenges faced by investors, rather than genuine risk premia or mispricing.

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- N assets, J firm characteristics, feature matrix $\mathbf{X}$ (size $N \times J$):
  - $\text{rank}(\mathbf{X}) = J$ (no redundant features, and assume $J < N$).
  - Standardized: $\frac{1}{NJ}\text{tr}(\mathbf{X}^\top \mathbf{X}) = 1$. (Let's see what this means.)

- Dividend growth model:
$$\Delta \boldsymbol{y}_t = \mathbf{X}\boldsymbol{g} + \boldsymbol{e}_t, \quad \boldsymbol{e}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N).$$

- Prior distribution of $\boldsymbol{g}$:
$$\boldsymbol{g} \sim \mathcal{N}\left(\mathbf{0}, \frac{\theta}{J}\mathbf{I}_J\right).$$
  Why do we need $1/J$ in the covariance? This is to ensure the signal-to-noise ratio does not change with J. (Why?)

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$$\Delta \boldsymbol{y}_t = \mathbf{X}\boldsymbol{g} + \boldsymbol{e}_t, \quad \boldsymbol{e}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N), \quad \mathbf{g} \sim \mathcal{N}\left(\mathbf{0}, \frac{\theta}{J}\mathbf{I}_J\right).$$

- Compared to real-world learning problems, the model includes several simplifications:
  - The set of relevant covariates is known.
  - The relationship between covariates and cash flows is linear and known.
  - The stochastic process driving fundamentals is time-invariant.
  - Investors optimally process all available information using Bayesian regression.

- These assumptions minimize estimation errors in learning cash flows.

- Real-world challenges:
  - Nonlinear relationships and time-varying features.
  - Large number of covariates ($J \geq N$).

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- Investors in the model are:
  - Homogeneous and risk-neutral. (Heterogeneity could reflect differences in data analysis methods and/or observed data, i.e., covariates.)
  - Operating under the assumption of zero interest rates.

- Implications:
  - Predictability of returns is not driven by risk premia or time-varying interest rates.
  - No joint hypothesis problem since there is no unknown risk pricing model.

- Focus:
  - How learning affects the interpretation of standard market efficiency tests.

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- We focus on single-period dividend claims:
  - Price vector $\boldsymbol{p}_t$: price at t for assets at $t+1$.

- Under the risk-neutral investor and zero interest rate assumptions, the pricing formula satisfies:

$$\boldsymbol{p}_t = \frac{\widetilde{\mathbb{E}}_t[\boldsymbol{y}_{t+1}]}{1 + \underbrace{R_f}_{=0} + \underbrace{\text{risk compensation}}_{=0}} = \widetilde{\mathbb{E}}_t[\boldsymbol{y}_{t+1}]$$

$$= \boldsymbol{y}_t + \widetilde{\mathbb{E}}_t[\Delta \boldsymbol{y}_{t+1}] = \boldsymbol{y}_t + \widetilde{\mathbb{E}}_t[\mathbf{X}\boldsymbol{g} + \boldsymbol{e}_{t+1}].$$

- How do investors form expectations $\widetilde{\mathbb{E}}_t[\cdot]$?

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Benchmark: Pricing under Rational Expectations

- Investors know the true $\boldsymbol{g}$.
- Rational expectations:

$$\widetilde{\mathbb{E}}_t[\mathbf{X}\boldsymbol{g} + \boldsymbol{e}_{t+1}] = \mathbf{X}\boldsymbol{g}.$$

- Price satisifies $\boldsymbol{p}_t = \boldsymbol{y}_t + \mathbf{X}\boldsymbol{g}$, and realized return ("yield") is:

$$\boldsymbol{r}_{t+1} = \boldsymbol{y}_{t+1} - \boldsymbol{p}_t = \Delta\boldsymbol{y}_{t+1} - \mathbf{X}\boldsymbol{g} = \boldsymbol{e}_{t+1}.$$

- $\boldsymbol{r}_{t+1} = \boldsymbol{e}_{t+1}$ means that returns are **unpredictable** under **rational expectations**.

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- An econometrician observes realized returns and conducts standard predictability tests, i.e., regressing realized returns $\mathbf{r}_{t+1}$ on firm characteristics $\mathbf{X}$.

- Regression coefficient:

$$\boldsymbol{h}_{t+1} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \boldsymbol{r}_{t+1}.$$

- Under rational expectations:

$$\begin{aligned} \boldsymbol{r}_{t+1} &= \boldsymbol{e}_{t+1}, \\ \boldsymbol{h}_{t+1} &= \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \boldsymbol{e}_{t+1}. \end{aligned}$$

- Deviations of $\boldsymbol{h}_{t+1}$ from zero are purely due to noise.

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- Distribution of $\boldsymbol{h}_{t+1}$ under rational expectations:

$$\sqrt{N}\boldsymbol{h}_{t+1} \sim \mathcal{N}\left(\mathbf{0}, N(\mathbf{X}^\top\mathbf{X})^{-1}\right).$$

- To test whether all elements of $\boldsymbol{h}_{t+1}$ are jointly zero, construct the following chi-squared test statistic:

$$\boldsymbol{h}_{t+1}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{h}_{t+1} \sim \chi_J^2.$$

- It can be tested using actual sample data. From the definition of $\boldsymbol{h}_{t+1}$, under rational expectations, any deviation from zero is caused by the noise $\boldsymbol{e}_{t+1}$.

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- Let $\boldsymbol{w}_t = (1/N)\mathbf{X}\boldsymbol{h}_{t+1}$, and consider a portfolio constructed using $\boldsymbol{w}_t$ as weights. This corresponds to the common practice of constructing portfolios in-sample and testing their returns.

- The return of this portfolio is:

$$r_{\text{IS},t+1} = \boldsymbol{w}_t^\top \boldsymbol{r}_{t+1} = \frac{1}{N}\boldsymbol{h}_{t+1}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{h}_{t+1}.$$

  Given that $\boldsymbol{h}_{t+1}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{h}_{t+1} \sim \chi^2_J$, the expected return of this portfolio is:

$$\mathbb{E}[r_{\text{IS},t+1}] = \frac{J}{N}.$$

  Under the null hypothesis of no predictability, the expected in-sample return of this portfolio is J/N. Why is the in-sample return not zero, but greater than zero?

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- In-sample return is greater than zero due to overfitting to in-sample noise:

  The overfitting arises because, in finite samples, random noise e and predictors $\mathbf{X}$ can exhibit spurious correlations that OLS exploits to minimize in-sample residuals.

- In traditional *ex post* tests, the usual approach is to examine whether the return of this portfolio significantly deviates from J/N.
  - If a significant deviation is found, it is often concluded that there exists predictability between $\mathbf{X}$ and $\boldsymbol{r}_{t+1}$, attributing this predictability to risk compensation or systematic biases of investors.

- What happens if investors do **NOT** know the true $\boldsymbol{g}$ and need to estimate it (**learning**)? Could estimation error lead to the null hypothesis being incorrectly rejected, i.e., spurious in-sample predictability?

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Investor Learning – Setup

- Investors use historical data to learn the parameter $\boldsymbol{g}$.
- Assume a known linear relationship between firm characteristics $\mathbf{X}$ and cash flow growth:

$$\Delta \boldsymbol{y}_t = \mathbf{X}\boldsymbol{g} + \boldsymbol{e}_t.$$

- Investors estimate $\boldsymbol{g}$ using linear regression.
- Key question: How does learning affect investor beliefs and asset prices?

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- Investors estimate $\boldsymbol{g}$ using OLS:

$$\tilde{\boldsymbol{g}}_{\text{OLS},t} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \overline{\Delta \boldsymbol{y}}_t, \quad \text{where} \quad \overline{\Delta \boldsymbol{y}}_t = \frac{1}{t} \sum_{s=1}^{t} \Delta \boldsymbol{y}_s.$$

- Price under OLS learning is $\boldsymbol{p}_t = \boldsymbol{y}_t + \mathbf{X}\tilde{\boldsymbol{g}}_{\text{OLS},t}$.

- Realized return:

$$\begin{aligned} \boldsymbol{r}_{t+1} &= \boldsymbol{y}_{t+1} - \boldsymbol{p}_t \\ &= \boldsymbol{y}_t + \mathbf{X}\boldsymbol{g} + \boldsymbol{e}_{t+1} - \boldsymbol{y}_t - \mathbf{X}\tilde{\boldsymbol{g}}_{\text{OLS},t} \\ &= \mathbf{X}(\boldsymbol{g} - \tilde{\boldsymbol{g}}_{\text{OLS},t}) + \boldsymbol{e}_{t+1}. \end{aligned}$$

- For investors at time t, they are unable to detect any bias in the estimation of $\boldsymbol{g}$. Therefore, for investors, $\boldsymbol{r}_{t+1}$ remains unpredictable. However, for *ex post* statistical testing, does the $\tilde{\boldsymbol{g}}_{\text{OLS},t}$ affect the test results?

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Impact on Realized Returns (OLS)

- Define $\bar{\boldsymbol{e}}_t = \frac{1}{t}\sum_{s=1}^{t} \boldsymbol{e}_s$, and realized returns become:

$$\begin{aligned} \boldsymbol{r}_{t+1} &= \mathbf{X}\left(\boldsymbol{g} - \tilde{\boldsymbol{g}}_{\text{OLS},t}\right) + \boldsymbol{e}_{t+1} \\ &= \mathbf{X}\left(\boldsymbol{g} - \boldsymbol{g} - \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\bar{\boldsymbol{e}}_t\right) + \boldsymbol{e}_{t+1} \\ &= -\mathbf{X}\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\bar{\boldsymbol{e}}_t + \boldsymbol{e}_{t+1}, \end{aligned}$$

- Let's compare:

Case	Realized Returns
Rational Expectation	$\boldsymbol{r}_{t+1} = \boldsymbol{e}_{t+1}$
OLS Learning	$\boldsymbol{r}_{t+1} = \underbrace{-\mathbf{X}\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\bar{\boldsymbol{e}}_t}_{\text{due to learning}} + \boldsymbol{e}_{t+1}$

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- *Ex post* regression coefficients:

$$\boldsymbol{h}_{t+1} = \underbrace{-\left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \bar{\boldsymbol{e}}_t}_{\text{due to OLS learning}} + \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \boldsymbol{e}_{t+1}.$$

- From the econometrician's perspective, **realized returns will appear predictable** if some firm characteristics in $\mathbf{X}$ are positively or negatively correlated with the error term $\bar{\boldsymbol{e}}_t$.

Predictability arises from estimation error in $\tilde{g}_{\mathbf{OLS},t}$ due to noise in historical cash flow growth.

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In-Sample Portfolio Returns

- Portfolio weights:

$$\boldsymbol{w}_t = \frac{1}{N}\mathbf{X}\boldsymbol{h}_{t+1}.$$

- Portfolio return:

$$r_{\text{IS},t+1} = \frac{1}{N}\boldsymbol{h}_{t+1}^{\top}\mathbf{X}^{\top}\mathbf{X}\boldsymbol{h}_{t+1}.$$

- Expected return:

$$\mathbb{E}[r_{\text{IS},t+1}] = \frac{J}{tN} + \frac{J}{N}.$$

- In high-dimensional settings ($J \sim N$), expected returns significantly exceed the rational expectations benchmark.

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- Investors do not know $\boldsymbol{g}$, but instead estimate $\boldsymbol{g}$ through OLS.

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Logic Flow

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- Investors do not know $\boldsymbol{g}$, but instead estimate $\boldsymbol{g}$ through OLS.
- They value assets based on $\tilde{\boldsymbol{g}}_{\mathrm{OLS},t}$, leading to **the equilibrium price of assets.**

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Logic Flow

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- This additional term further causes the *ex post* regression coefficient $\boldsymbol{h}_{t+1}$ to significantly deviate from zero, or the expected portfolio return to significantly deviate from J/N, leading econometrician to **incorrectly reject** the null hypothesis.

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- This, in turn, results in an additional term in realized returns compared to rational expectations.
- This additional term further causes the *ex post* regression coefficient $\boldsymbol{h}_{t+1}$ to significantly deviate from zero, or the expected portfolio return to significantly deviate from J/N, leading econometrician to **incorrectly reject** the null hypothesis.

Predictability is an artifact of the learning process, not risk premia or biases.

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- OLS is not necessarily optimal. Predictability may reflect suboptimal learning methods rather than fundamental market inefficiencies.
- Investors use Bayesian updating to learn $\boldsymbol{g}$.
- To focus on "learning" (i.e., estimation), assume that investors' prior belief about $\boldsymbol{g}$ is the true prior distribution:

$$\boldsymbol{g} \sim \mathcal{N}\left(\mathbf{0}, \frac{\theta}{J}\mathbf{I}_J\right).$$

- θ determines the prior variance and the degree of shrinkage.

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- Posterior mean of $\boldsymbol{g}$:

$$\tilde{\boldsymbol{g}}_t = \left(\mathbf{X}^\top \mathbf{X} + \frac{J}{\theta t}\mathbf{I}_J\right)^{-1} \mathbf{X}^\top \overline{\Delta \boldsymbol{y}}_t.$$

- Using eigen-decomposition $\frac{1}{N}\mathbf{X}^\top\mathbf{X} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$, rewrite $\tilde{\boldsymbol{g}}_t$ as a shrinkage estimator:

$$\tilde{\boldsymbol{g}}_t = \boldsymbol{\Gamma}_t \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \overline{\Delta \mathbf{y}}_t, \quad \text{where} \quad \boldsymbol{\Gamma}_t = \mathbf{Q}\left(\mathbf{I}_J + \frac{J}{N\theta t}\boldsymbol{\Lambda}^{-1}\right)^{-1}\mathbf{Q}^\top.$$

- Shrinkage effect:
  - Stronger shrinkage for small eigenvalues of $\boldsymbol{\Lambda}$.
  - Shrinkage increases when t or θ is small, or when J/N is large.

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- Shrinkage prevents overly large $\boldsymbol{g}$ estimates, which are economically implausible:
  - The smaller t, the more it shrinks toward the prior (shorter sample time span leads to larger errors);
  - The smaller θ, the more it shrinks toward the prior (θ determines the extent to which $\boldsymbol{g}$ deviates from zero in the prior);
  - The larger J/N, the more it shrinks toward the prior (the more variables there are, the more likely it is to overfit the sample, thus requiring more shrinkage).

- When the prior is diffuse ($\theta \to \infty$), $\boldsymbol{\Gamma}_t$ converges to the identity matrix, and OLS learning is a special case of Bayesian learning.

- OLS lacks economic rationality in high-dimensional settings ($J \sim N$). Bayesian learning incorporates prior information to achieve more realistic and stable estimates.

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- Under Bayesian learning, investors assess dividend growth using $\tilde{\boldsymbol{g}}_t$ and value assets accordingly. In equilibrium, the returns satisfy:

$$\boldsymbol{r}_{t+1} = \mathbf{X}\left(\mathbf{I}_J - \boldsymbol{\Gamma}_t\right)\boldsymbol{g} - \mathbf{X}\boldsymbol{\Gamma}_t\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\bar{\boldsymbol{e}}_t + \boldsymbol{e}_{t+1}.$$

- Let's compare:

Case	Realized Returns
Rational Expectation	$\boldsymbol{r}_{t+1} = \boldsymbol{e}_{t+1}$
OLS Learning	$\boldsymbol{r}_{t+1} = -\mathbf{X}\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\bar{\boldsymbol{e}}_t + \boldsymbol{e}_{t+1}$
Bayesian Learning	$\boldsymbol{r}_{t+1} = \mathbf{X}\left(\mathbf{I}_J - \boldsymbol{\Gamma}_t\right)\boldsymbol{g} - \mathbf{X}\boldsymbol{\Gamma}_t\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\bar{\boldsymbol{e}}_t + \boldsymbol{e}_{t+1}$

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$$\boldsymbol{r}_{t+1} = \mathbf{X}\left(\mathbf{I}_J - \boldsymbol{\Gamma}_t\right)\boldsymbol{g} - \mathbf{X}\boldsymbol{\Gamma}_t\left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\bar{\boldsymbol{e}}_t + \boldsymbol{e}_{t+1}.$$

- The first term arises due to shrinkage toward the prior, leading to investors' "underreaction" to the fundamental information $\mathbf{X}$ (without shrinkage, i.e., $\boldsymbol{\Gamma}_t = \mathbf{I}$, this term will disappear).

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- The last term, like in rational expectations, corresponds to $\boldsymbol{e}_{t+1}$.

Shrinkage reduces noise-induced errors but introduces underreaction to fundamentals. $\boldsymbol{\Gamma}_t$ achieves an optimal balance between the first two terms.

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- *Ex post* regression coefficients:

$$\boldsymbol{h}_{t+1} = (\mathbf{I}_J - \boldsymbol{\Gamma}_t)\,\boldsymbol{g} - \boldsymbol{\Gamma}_t \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \bar{\boldsymbol{e}}_t + \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \boldsymbol{e}_{t+1}.$$

- Expected return of in-sample portfolio:

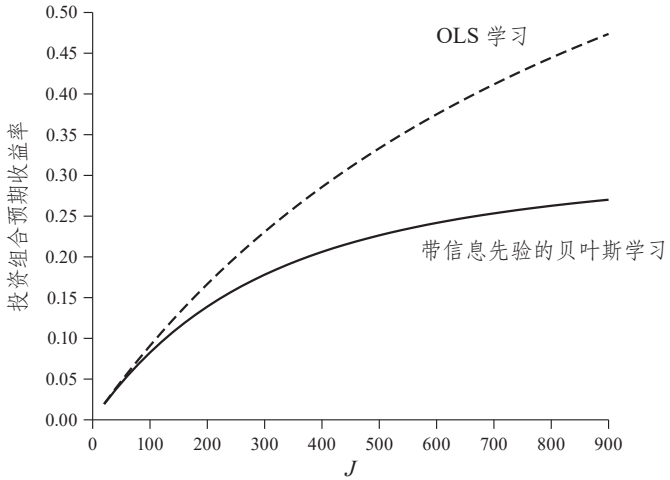
$$\mathbb{E}[r_{\mathrm{IS},t+1}] = \frac{1}{N} \sum_{j=1}^{J} \left(\frac{\lambda_j}{t\lambda_j + \frac{J}{N\theta}} + 1 \right).$$

- When $\theta \to \infty$, it converges to the case of OLS learning, i.e., $\mathbb{E}[r_{\mathrm{IS},t+1}] = \frac{J}{tN} + \frac{J}{N}$.

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Illustration: In-Sample Predictability

- When there is no prior, OLS learning is more prone to overfitting, causing its expected return to increase more rapidly with J.
- With Bayesian shrinkage, this situation is mitigated to some extent but it cannot fully eliminate in-sample predictability.



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Logic Flow (One More Time)

- Investors do not know $\boldsymbol{g}$, but instead estimate $\boldsymbol{g}$ through Bayesian learning.

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Logic Flow (One More Time)

- Investors do not know $\boldsymbol{g}$, but instead estimate $\boldsymbol{g}$ through Bayesian learning.
- They value assets based on $\tilde{\boldsymbol{g}}_t$, leading to **the equilibrium price of assets.**

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Even with the more realistic Bayesian learning, investors' high-dimensional learning still generates false in-sample predictability.

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- In-sample predictability arises from using hindsight information unavailable to investors.
- Out-of-sample (OOS) predictability tests place econometricians and investors on equal footing.
- OOS portfolio return is defined as:

$$r_{\text{OOS},t+1} = \boldsymbol{w}_{\text{OOS},t}^{\top} \boldsymbol{r}_{t+1} \quad \text{with} \quad \boldsymbol{w}_{\text{OOS},t} = \frac{1}{N} \mathbf{X} \boldsymbol{h}_t.$$

  This strategy uses only information available up to time t.
- $\mathbb{E}[r_{\text{OOS},t+1}] = 0$, even in high-dimensional learning environments.

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Implications for Empirical Research

- High-dimensional learning introduces in-sample predictability but not OOS predictability.
- Focus should shift from in-sample predictability to OOS predictability when assessing risk premia or behavioral biases.
- **Reverse OOS tests** highlight that many cross-sectional anomalies fail under this framework (Linnainmaa and Roberts 2018).
- This suggests that anomalies may stem from high-dimensional learning errors rather than risk premia or behavioral biases. (Implications for alternative data?)

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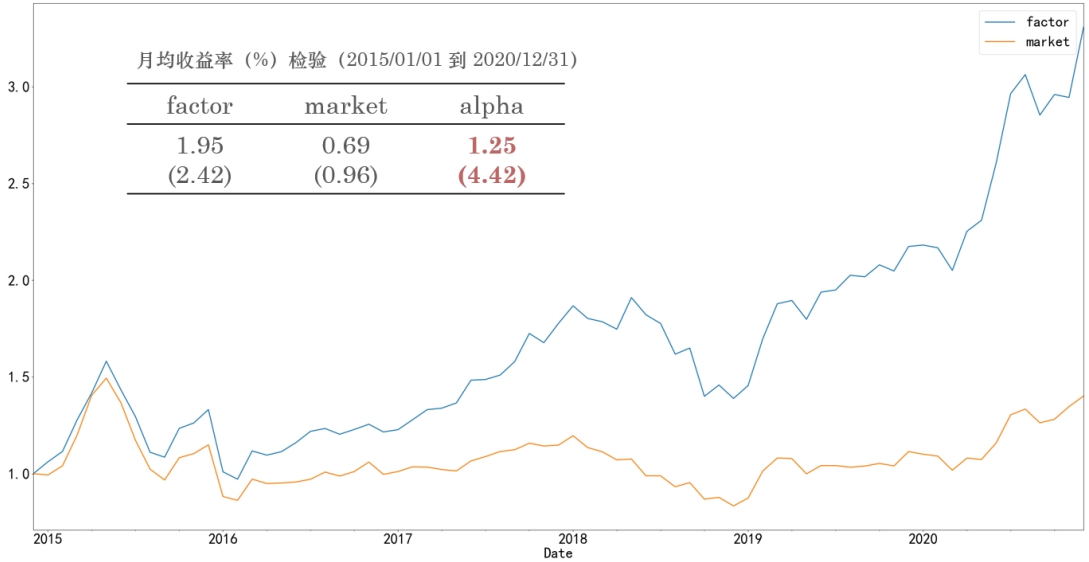
- OOS = Pre-Sample + Post-Sample.
- False factors: No excess returns in both pre-sample and post-sample periods.
- True factors: More significant in the pre-sample period.

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Example 1: In-Sample

月均收益率（%）检验（2015/01/01 到 2020/12/31）

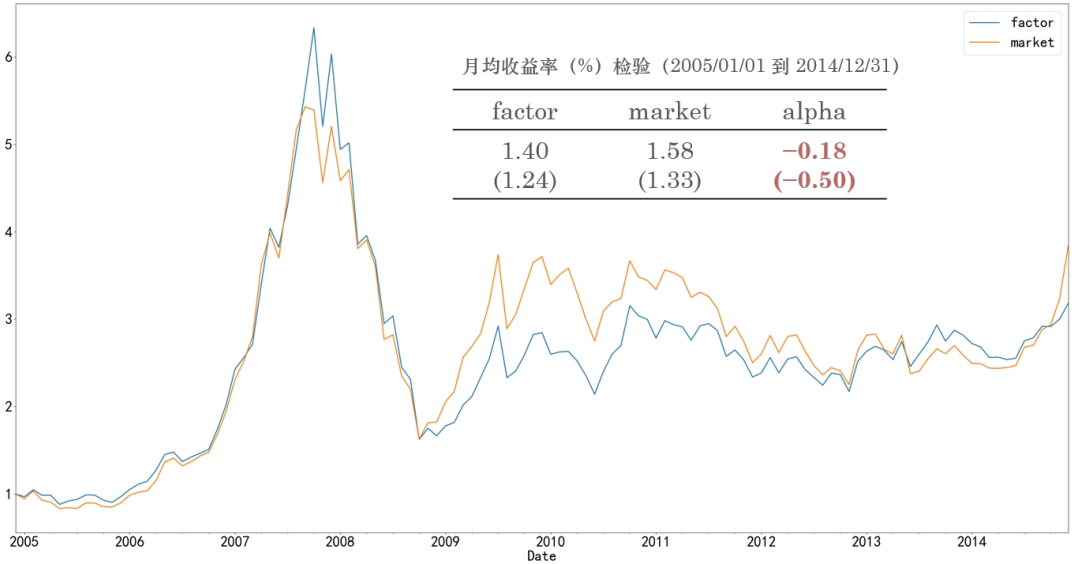
factor	market	alpha
1.95	0.69	**1.25**
(2.42)	(0.96)	**(4.42)**



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Example 1: Pre-Sample

3 Investor Learning



月均收益率（%）检验（2005/01/01 到 2014/12/31）

factor	market	alpha
1.40	1.58	**−0.18**
(1.24)	(1.33)	**(−0.50)**

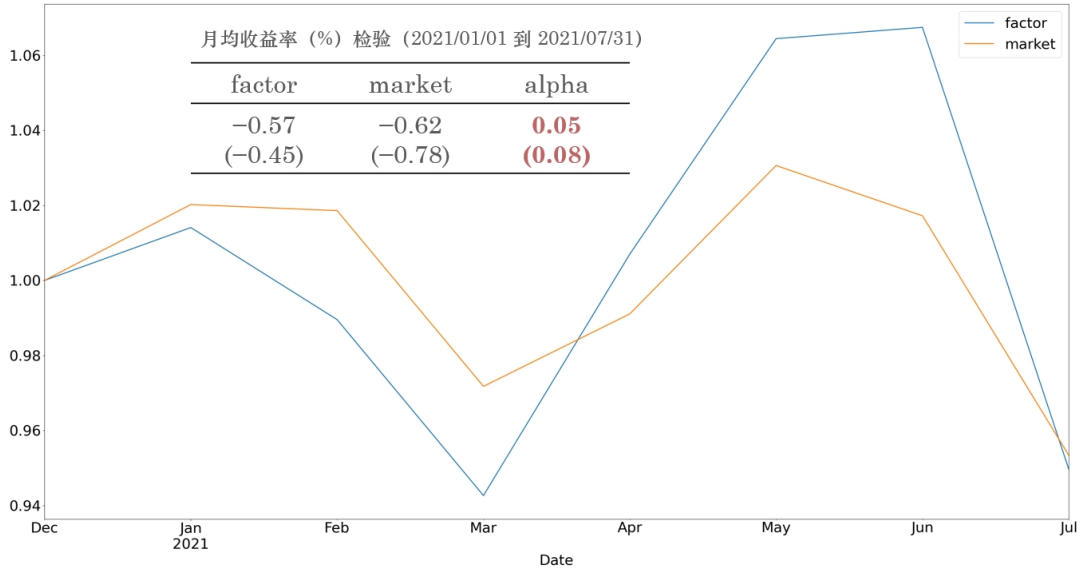
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Example 1: Post-Sample

3 Investor Learning

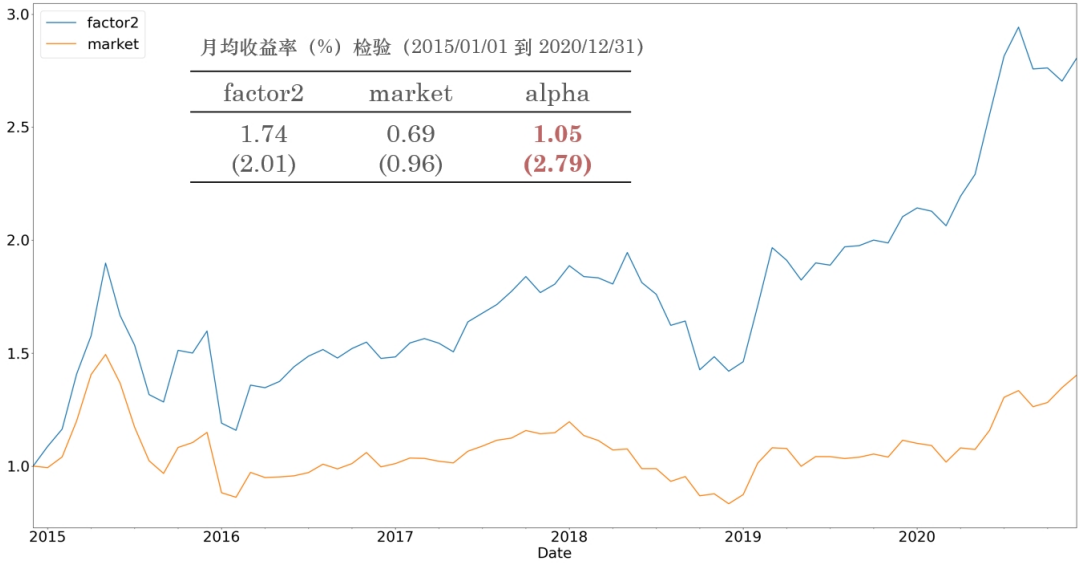
月均收益率（%）检验（2021/01/01 到 2021/07/31）

factor	market	alpha
−0.57	−0.62	**0.05**
(−0.45)	(−0.78)	**(0.08)**



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Example 2: In-Sample



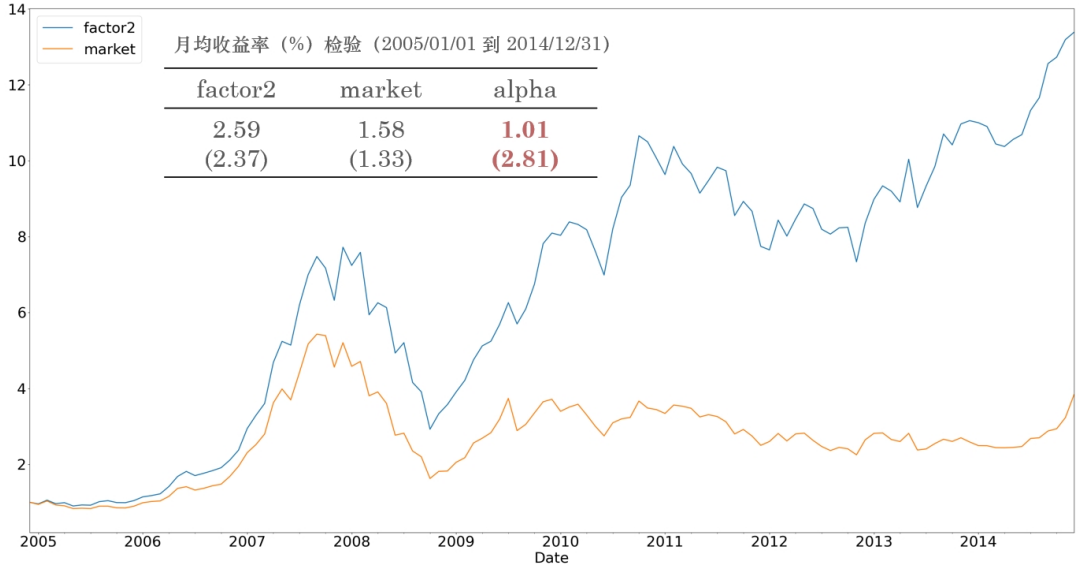
月均收益率（%）检验（2015/01/01 到 2020/12/31）

factor2	market	alpha
1.74	0.69	**1.05**
(2.01)	(0.96)	**(2.79)**

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Example 2: Pre-Sample

3 Investor Learning

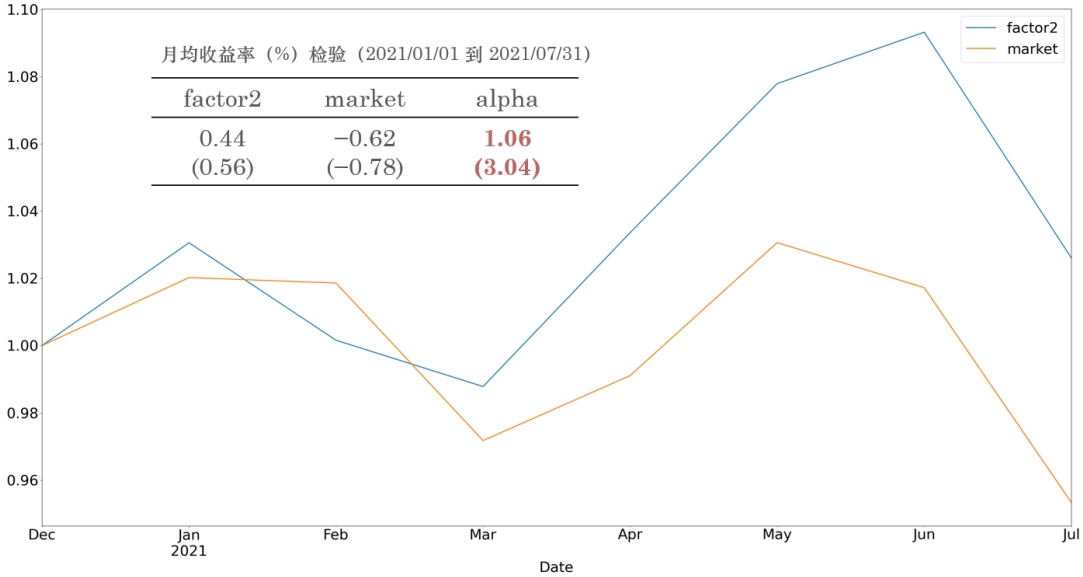


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Example 2: Post-Sample

月均收益率（%）检验（2021/01/01 到 2021/07/31）

factor2	market	alpha
0.44	−0.62	**1.06**
(0.56)	(−0.78)	**(3.04)**



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Questions?

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